

SMART INDIA HACKATHON 2026



SETU - Sensor-Enabled Tracking of Urban-road-damage

Millions of phones are already on the road. SETU puts them on work for road-condition check

Problem Statement ID: SIH 2026 - Smart Automation (to be filled)

Problem Statement Title: Crowdsourced Road Defect Intelligence for India

Theme: Smart Automation

PS Category: Software

Team ID: To be filled

Team Name: Team SETU

9,438

POTHOLE DEATHS

2020-24 (MoRTH), up 53% in 5 years

1.2 Cr

GIG-WORKER PHONES

already on the road, 12 h a day

Rs 3

PER CONFIRMED POTHOLE

a manual audit costs Rs 17,693

0.1%

OF DATA NEEDS THE CAMERA

sensors do the rest for free



THE PROBLEM WE ARE SOLVING

From 2020 to 2024, potholes were linked to 9,438 deaths on Indian roads, and the count has gone up 53% in five years. Cities still find bad patches the slow way: a survey van that passes once or twice a year, or a citizen complaint that waits weeks for someone to visit. So nobody actually has a live, trustworthy map of where the potholes are - and you cannot repair what you cannot see.

9,438 deaths in 5 years

up 53% since 2019

Rs 17,693 to audit one pothole

1-2 checks a year

HOW IT WORKS: CHEAP FIRST, CAMERA ONLY IF NEEDED

1

The phone feels the bump

A driver's phone is already mounted on the handlebar or dashboard. We read its motion sensors and a small model on the phone decides if that jolt was a real road defect. The driver does nothing.

Rs 0

per reading

100%

of roads driven

only if 3+ vehicles report the same spot

2

Several vehicles have to agree

Our server groups readings that land on the same stretch of road. A spot is only taken seriously once three or more separate vehicles have reported it.

Rs 0

per reading

top 5%

spots kept

only if the sensor data is unclear

3

The camera confirms the doubtful ones

When the sensor data is not clear enough, the next driver passing that spot records a 5-8 second clip and our vision model checks what is actually on the road.

Rs 3

per clip

0.1%

of spots

WHAT MAKES IT DIFFERENT

- Nothing new to install**

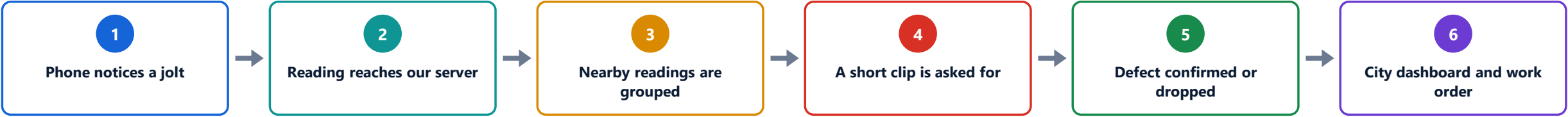
It sits inside apps drivers already run all day, so there is no extra device to install or carry.
- A repair proves itself**

When a fixed road stops sending jolts, the ticket closes automatically. Silence is the proof.
- Every vehicle reads differently, and we handle it**

Each phone first learns its own baseline, so readings from bikes, autos and buses can be compared more reliably.
- Built to stay private**

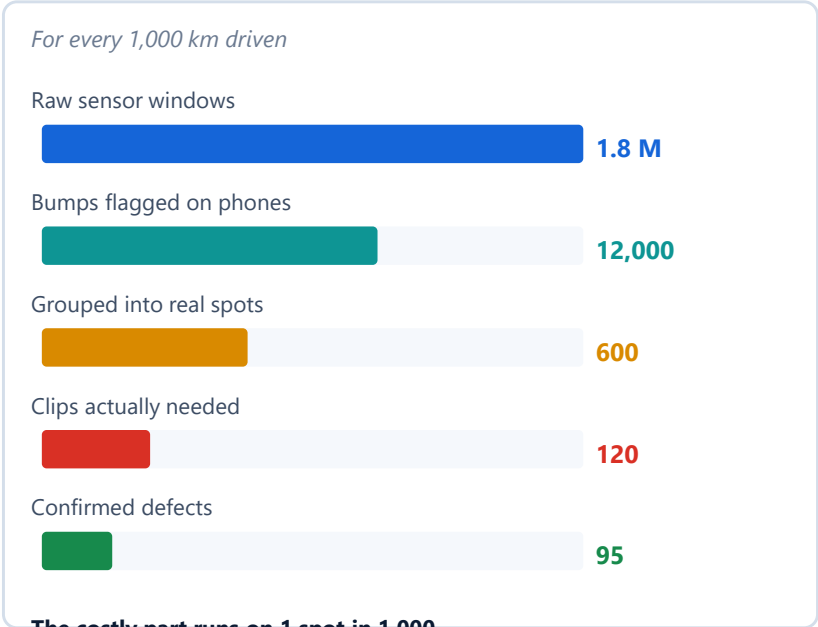
No personal details are collected, device IDs are hashed and clips are deleted after seven days.

HOW THE DATA MOVES, START TO FINISH



The system learns from the outcome: repaired locations stop generating alerts, while confirmed and rejected cases can be added to the training data.

WHY THE MATH WORKS



WHAT WE ARE BUILDING IT WITH

Driver app

- React Native
- TensorFlow Lite
- Android foreground service
- Room / SQLite
- Fused Location
- CameraX

Backend

- Python FastAPI
- Celery workers
- Redis Streams
- WebSockets
- JWT auth
- OSRM map-matching

Data

- PostgreSQL 16
- PostGIS
- H3 grid
- DBSCAN (scikit-learn)
- S3 object storage

Web dashboard

- React + TypeScript
- Tailwind CSS
- deck.gl
- MapLibre GL
- Recharts
- Socket.IO
- react-i18next

Models and training

- YOLOv11s (Ultralytics, PyTorch)
- 1D-CNN (Keras)
- RDD2022 + our Indian data
- CVAT labelling
- INT8 quantisation
- Colab / RTX GPU

Running it

- Docker Compose
- Nginx
- GitHub Actions
- Grafana + logs
- IRC:82 severity rules
- DPDP-safe storage

FEASIBILITY AND VIABILITY

IS IT ACTUALLY DOABLE?

CAN WE BUILD IT

- Every piece is free and open source, and all of it is already widely used
- Runs on an ordinary Rs 8,000 Android phone, no new hardware
- We have already started collecting and cleaning our training images
- Six of us, and the plan is split into parts we can each own

WILL IT RUN DAY TO DAY

- The system runs in apps drivers already use
- One city bus route or municipal fleet is enough to start
- Works with no network and sends data later on Wi-Fi
- A ward officer only has to read a list and assign work

DOES THE MONEY ADD UP

- About Rs 3 per confirmed pothole against Rs 17,693 for a manual audit
- No hardware to buy: the phones and the vehicles already exist
- Free cloud tiers are enough for the whole prototype
- Costs grow with confirmed defects, not with kilometres watched

IS IT ALLOWED

- No personal data, video only with consent, kept for seven days
- Every detection leaves a record, so an RTI query can be answered
- Severity follows the IRC:82 road maintenance code
- Data can be handed to the city as open GeoJSON

WHAT COULD GO WRONG, AND OUR ANSWER

Phones kill background apps - Xiaomi and Oppo shut down anything running quietly

Run it as a visible service the user can see, ask once for autostart, and test on MIUI and ColorOS

GPS can be off by 20-30 m - at speed the location drifts

Snap every reading to the road with OSRM, then group points within a 20 m radius

A bike and a bus feel a bump differently - same pothole, very different shake

Ten minutes of self-calibration per phone, and we adjust the reading for speed

Speed breakers look like potholes - so do manholes and expansion joints

We train on those on purpose as 'not a pothole', need three vehicles to agree, and let the camera overrule

Privacy, and poor light at night - video can catch faces and plates

Faces and plates are blurred on the phone, clips expire in seven days, and sensors carry the night shift

For SIH we will show one Android app, one backend and one dashboard, tested on about 50 km of real city road.

WHAT CHANGES ON THE GROUND

6+
DEATHS A DAY
pothole crashes, MoRTH

+53%
RISE IN 5 YEARS
and still climbing

Daily
ROAD CHECK
instead of once in 6-12 months

100%
REPAIRS CHECKABLE
evidence before and after

Who	Today	With SETU
Citizens	no warning, nobody answerable	safer roads, repairs on a public clock
Municipality	costly audits, guesswork	a ranked repair list, updated daily
NHAI / PWD	one survey pass a year	the whole network checked every day
Gig workers	vehicle damage, back pain	warnings ahead, no extra work
Researchers	no open Indian road data	a shared sensor and video dataset

5,900x

cheaper for the same proof
Rs 17,693 for one pothole in a manual audit against about Rs 3 with sensors first and the camera only when needed.

THE COST AND THE COVERAGE

WHAT IT COSTS TO CONFIRM ONE POTHOLE

Survey vehicle or manual audit

Rs 17,693

Citizen complaint app

cannot be verified

SETU

Rs 3

Same kind of evidence, a fraction of the money.

HOW OFTEN A ROAD GETS CHECKED IN A YEAR

Survey vehicle

once or twice, highways only

SETU

every trip, on every lane driven

SIDE BENEFITS WE DID NOT PLAN FOR

- Monsoon damage shows up in days instead of at the next survey
- Smoother roads mean less fuel burnt and fewer suspension repairs
- The same data tells a city which contractor actually delivered

PEER-REVIEWED FOUNDATION (each line = what it proves for SETU)

Sattar, Li & Chapman - Sensors 18:3845 (2018)

Review proving a smartphone IMU can detect road anomalies; sampling rate is the real limit. Hence, we sample at 100 Hz.

mdpi.com/1424-8220/18/11/3845

Wu et al. - Sensors 20:5564 (2020)

Automated ML pothole detection from phone data: precision 88.5% / recall 75% - the baseline our edge model must beat.

mdpi.com/1424-8220/20/19/5564

Sensors 20:409 (2020) - participatory sensing

Measured GPS error of 27-32 m at speed, forces our DBSCAN eps=20 m + OSRM map-matching design.

mdpi.com/1424-8220/20/2/409

Arya, Maeda, Ghosh et al. - RDD2022 (arXiv)

47,420 annotated road images from 6 countries including India - the training base for our YOLOv11s vision layer.

arxiv.org/abs/2209.08538

GOVERNMENT DATA & INDIAN STANDARDS

MoRTH - Road Accidents in India

Official data: 9,438 pothole-related deaths in 2020-24, +53% in five years, several states reporting zero.

morth.nic.in/road-accident-in-india

NHAI Network Survey Vehicles

The survey fleet covers ~20,933 km of highways about once a year at high cost - SETU covers every lane, daily.

nhai.gov.in

IRC:82 - Maintenance of Bituminous Surfaces

The Indian Roads Congress code we map severity to (depth x width x traffic), so our output is directly actionable.

[law.resource.org - IRC 82:2015 PDF](https://law.resource.org/IRC%2082%202015%20PDF)

DPDP Act 2023 + DPDP Rules (13 Nov 2025)

Our privacy frame: no PII, hashed device tokens, consent-first capture, purpose limits, 7-day media TTL.

[meity.gov.in - data protection framework](https://meity.gov.in/data-protection-framework)